

Batch: A2 Roll No.: 16010324044

Experiment / assignment / tutorial No. 4

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Identify MAC and IP Addresses

AIM: To identify MAC and IP Addresses for given network topology

OUTCOME: Student will be able to perceive the significance of data link layer services and protocols.

Theory:

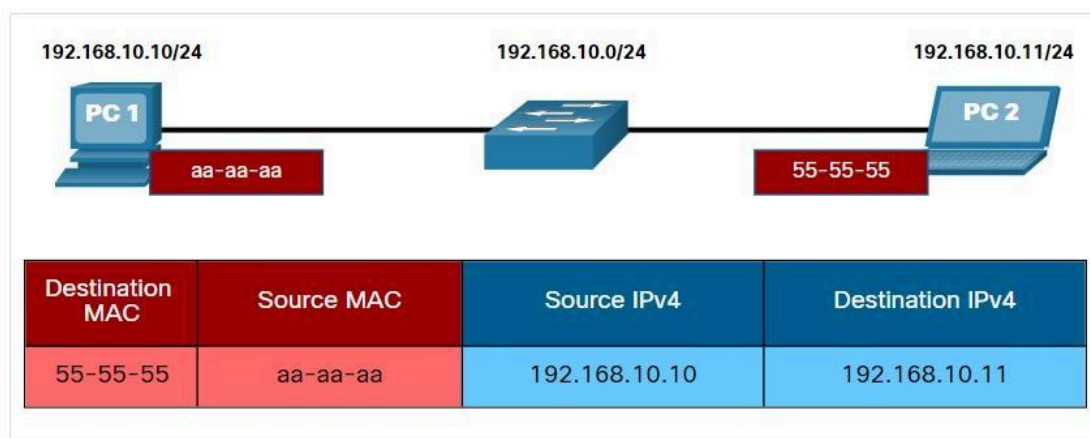
Sometimes a host must send a message, but it only knows the IP address of the destination device. The host needs to know the MAC address of that device, but how can it be discovered? That is where address resolution becomes critical.

There are two primary addresses assigned to a device on an Ethernet LAN:

- **Physical address (the MAC address)** – Used for NIC to NIC communications on the same Ethernet network.
- **Logical address (the IP address)** – Used to send the packet from the source device to the destination device. The destination IP address may be on the same IP network as the source or it may be on a remote network.

Layer 2 physical addresses (i.e., Ethernet MAC addresses) are used to deliver the data link frame with the encapsulated IP packet from one NIC to another NIC that is on the same network. If the destination IP address is on the same network, the destination MAC address will be that of the destination device.

Consider the following example using simplified MAC address representations.



In this example, PC1 wants to send a packet to PC2. The figure displays the Layer 2 destination and source MAC addresses and the Layer 3 IPv4 addressing that would be included in the packet sent from PC1.

The Layer 2 Ethernet frame contains the following:

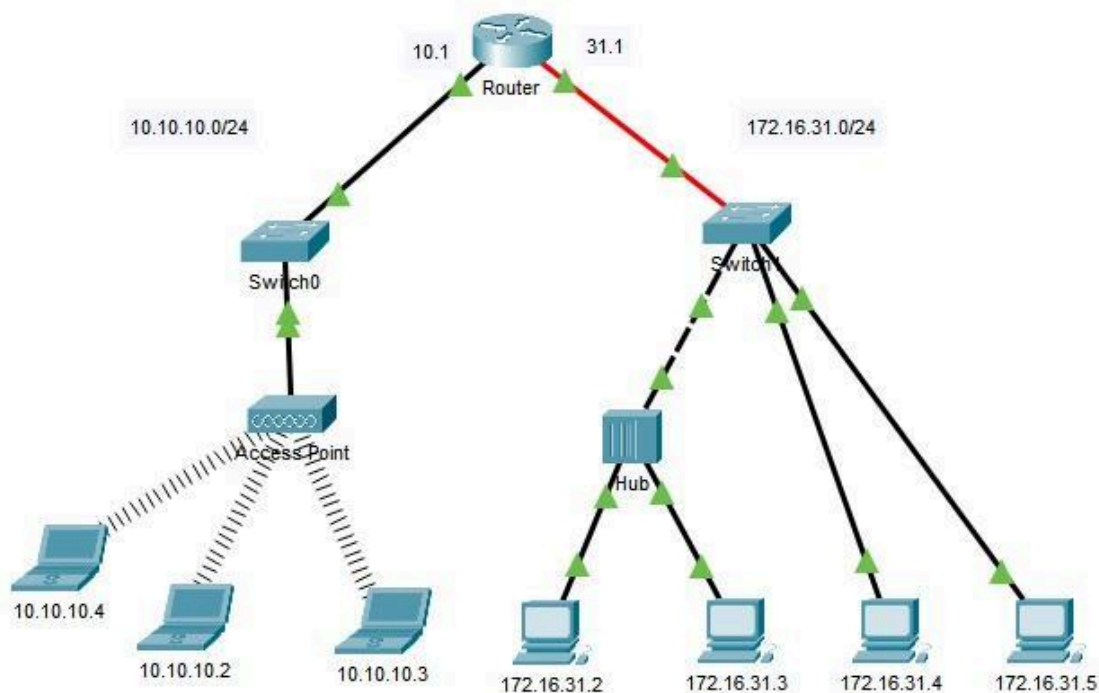
- **Destination MAC address** – This is the simplified MAC address of PC2, 55-55-55.
- **Source MAC address** – This is the simplified MAC address of the Ethernet NIC on PC1, aa-aa-aa.

The Layer 3 IP packet contains the following:

- **Source IPv4 address** – This is the IPv4 address of PC1, 192.168.10.10.
- **Destination IPv4 address** – This is the IPv4 address of PC2, 192.168.10.11.

Procedure:

1. Configure the topology as shown in figure below:



Part 1: Gather PDU Information for Local Network Communication

Step1: Identify the type of information you will need to gather PDU information as a packet travels from 172.16.31.5 to 172.16.31.2.

- Click **172.16.31.5** and open the **Command Prompt**.
- Enter the **ping 172.16.31.2** command.
- Switch to simulation mode and repeat the **ping 172.16.31.2** command. A PDU appears next to **172.16.31.5**.
- Click **Capture / Forward (the right arrow followed by a vertical bar)** to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered into a spreadsheet using a format like the table shown below:
- Click the PDU and note the following information from the **OSI Model** and **Outbound PDU Layer** tabs:

- o Destination MAC Address: **000C:85CC:1DA7**
- o Source MAC Address: **00D0:D311:C788**

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- o Source IP Address: **172.16.31.5**
- o Destination IP Address: **172.16.31.2**
- o At Device: **172.16.31.5**

Example Spreadsheet Format:

At Device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.5	000C:85CC:1DA7	00D0:D311:C788	172.16.31.5	172.16.31.2
Switch1	000C:85CC:1DA7	00D0:D311:C788	N/A	N/A
Hub	N/A	N/A	N/A	N/A
172.16.31.2	00D0:D311:C788	000C:85CC:1DA7	172.16.31.2	172.16.31.5

Step 2: Gather additional PDU information from other pings.

Repeat the process in Step 1 and gather the information for the following tests:

Ping 172.16.31.2 from 172.16.31.3

At device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.2	FFFF.FFFF.FF FF	0030.A3E6.C1 D2	172.16.31.2	172.16.31.3
Hub0	NA	NA	NA	NA
172.16.31.3	0030.A3E6.C1 D2	0001.C965.318 5	172.16.31.3	172.16.31.2

Ping 172.16.31.4 from 172.16.31.5.

At device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.4	FFFF.FFFF.FF FF	0001.6421.9D D8	172.16.31.4	172.16.31.5
Switch0	FFFF.FFFF.FF FF	0001.6421.9D D8	NA	NA
172.16.31.5	0001.6421.9D D8	0001.6421.9D D8	172.16.31.4	172.16.31.5

Return to Realtime mode.

Part 2: Gather PDU Information for Remote Network Communication

In order to communicate with remote networks, a gateway device is necessary. Study the process that takes place to communicate with devices on the remote network. Pay close attention to the MAC addresses used.

Step 1: Gather PDU information as a packet travels from 172.16.31.5 to 10.10.10.2.

- a. Click 172.16.31.5 and open the Command Prompt.
- b. Enter the ping 10.10.10.2 command.
- c. Switch to simulation mode and repeat the ping 10.10.10.2 command. A PDU appears next to 172.16.31.5.
- d. Click the PDU and note the following information from the Outbound PDU Layer tab:
 - Destination MAC Address: 00D0:BA8E:741A
 - Source MAC Address: 00D0:D311:C788
 - Source IP Address: 172.16.31.5
 - Destination IP Address: 10.10.10.2
 - At Device: 172.16.31.5

Question:

What device has the destination MAC that is shown?

The destination MAC address belongs to the Switch0 because we are at Source IP Device and will forward next to Switch.

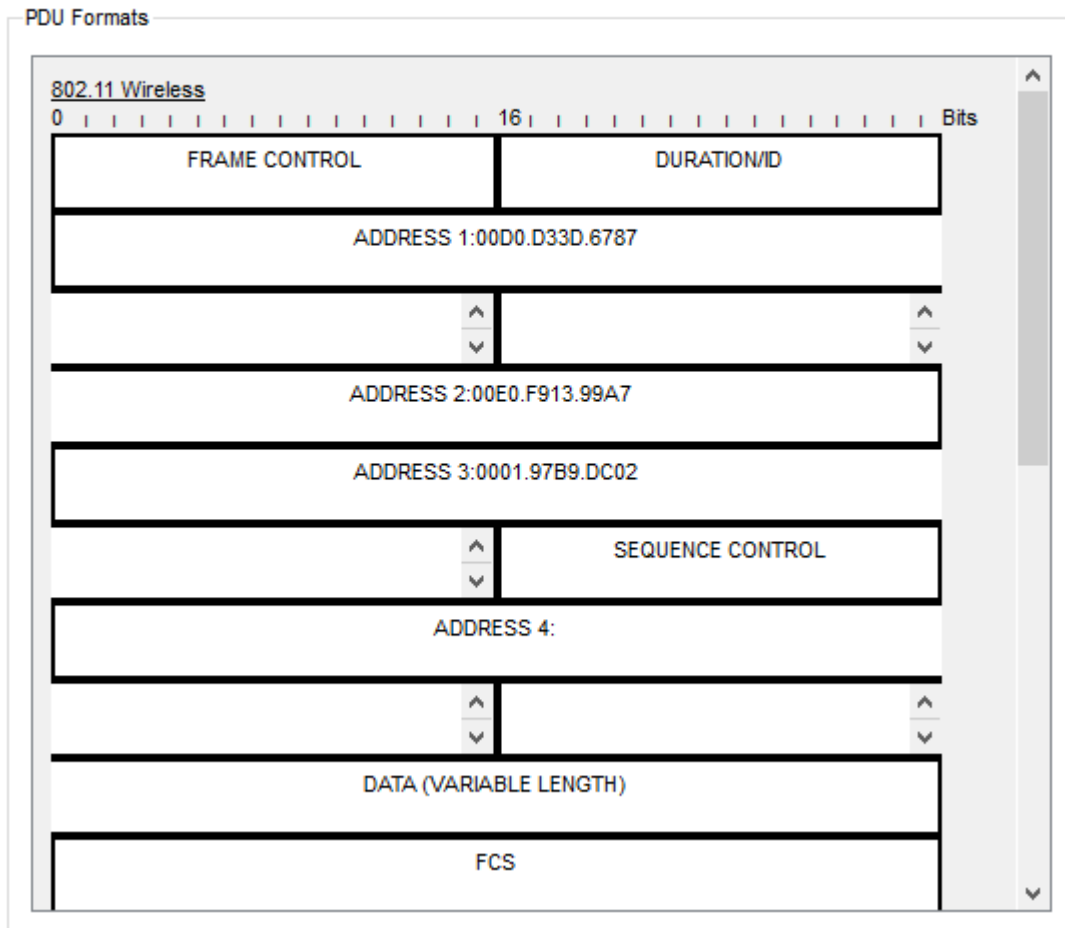
- e. Click Capture / Forward (the right arrow followed by a vertical bar) to move the PDU to the next device. Gather the same information from Step 1d. Repeat this process until the PDU reaches its destination. Record the PDU information you gathered from pinging 172.16.31.5 to 10.10.10.2 into a spreadsheet table shown below:

At device	Dest. MAC	Src MAC	Src IPv4	Dest IPv4
172.16.31.5	0001.97B9.DC 01	0007.EC17.211 8	172.16.31.5	10.10.10.2
Switch0	0001.97B9.DC 01	0007.EC17.211 8	NA	NA
Router0	00D0.D33D.67 87	0001.97B9.DC 02	172.16.31.5	10.10.10.2
Switch1	00D0.D33D.67 87	0001.97B9.DC 02	172.16.31.5	10.10.10.2
Access Point0	NA	NA	NA	NA
Laptop1	00D0.D33D.67 87	0001.97B9.DC 02	172.16.31.5	10.10.10.2

Observe the ARP packet and write down the source and destination address for the first packet transmission and second packet transmission

PDU Information at Device: Access Point0

OSI Model Inbound PDU Details Outbound PDU Details



CONCLUSION

In this experiment, I successfully configured the network topology in Cisco Packet Tracer and observed packet transmission in Simulation Mode. It was verified that MAC addresses are used for local (Layer 2) communication and change at each hop, whereas IP addresses are used for end-to-end (Layer 3) communication and remain unchanged from source to destination. ARP was also observed resolving IP addresses into MAC addresses before data transmission.

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